

CREDITPATHAI PROJECT DOCUMENTATION

1. Project Title

CreditPathAI - Automating and optimizing the loan recovery lifecycle by modeling repayment behavior using diverse data.

2. Abstract

CreditPathAI is an end-to-end machine learning-based application developed to predict the probability of loan default using borrower financial and demographic data. The system integrates advanced machine learning models with a modern web-based interface to provide real-time risk assessment and decision-making support.

The project combines:

- Machine Learning models (XGBoost & LightGBM)
- FastAPI backend for prediction serving
- React-based frontend dashboard for visualization

The application helps financial institutions classify borrowers into risk categories and take appropriate actions.

3. Objectives

- To predict loan default probability using ML models
- To classify borrowers into **Low, Medium, High Risk**
- To build a complete **full-stack AI system**
- To provide actionable insights for decision-making
- To create an interactive UI dashboard for visualization

4. System Architecture

The system consists of three main layers:

4.1 Machine Learning Layer

- Data preprocessing and cleaning
- Feature engineering

- Model training using:
 - XGBoost
 - LightGBM
- Handling class imbalance
- Probability calibration

4.2 Backend Layer (FastAPI)

- REST API for prediction
- Model loading (model.pkl, pipeline.pkl)
- Risk classification logic
- Asynchronous request handling

4.3 Frontend Layer (React + Vite)

- User input form
- Risk visualization (Gauge charts)
- KPI dashboard
- Local storage tracking

5. Project Structure

```
creditpathai/
├── frontend/
│   ├── src/
│   ├── index.html
│   ├── vite.config.js
│   └── package.json
├── main.py
├── utils.py
├── advanced_model_training.py
├── eda_analysis.py
├── model.pkl
├── pipeline.pkl
└── README.md
```

6. Technologies Used

Layer	Technology Used
Frontend	React, Vite, Tailwind CSS
Backend	FastAPI, Uvicorn

Layer	Technology Used
ML Libraries	Scikit-learn, XGBoost, LightGBM
Data Tools	Pandas, NumPy
Visualization	Plotly.js

7. Methodology

Step 1: Data Collection

- Dataset contains borrower information such as income, credit score, loan details.

Step 2: Data Preprocessing

- Handling missing values
- Feature scaling
- Encoding categorical variables

Step 3: Feature Engineering

- Creation of new features like:
 - Debt-to-Income Ratio (DTI)
 - Monthly interest burden

Step 4: Model Training

- Used ensemble models:
 - XGBoost
 - LightGBM
- Performance evaluated using ROC-AUC

Step 5: Model Deployment

- Saved trained model using `joblib`
- Loaded into FastAPI backend

Step 6: Frontend Integration

- React UI sends input data to API
- Displays prediction and risk category

8. Risk Classification Logic

Probability Range	Risk Level	Action
< 0.25	Low Risk	Regular follow-up
0.25 – 0.40	Medium Risk	Send reminder
≥ 0.40	High Risk	Immediate action

9. API Design

Endpoint:

POST /predict

Input:

```
{
  "age": 35,
  "income": 50000,
  "loan_amount": 10000,
  "credit_score": 750,
  "months_employed": 24,
  "num_credit_lines": 3,
  "interest_rate": 5.5,
  "loan_term": 36,
  "dti_ratio": 0.2
}
```

Output:

```
{
  "probability": 0.2752,
  "risk": "Medium Risk",
  "action": "Send reminder and monitor"
}
```

10. Key Features

10.1 Machine Learning

- Gradient boosting models
- Feature engineering pipeline
- Probability calibration

10.2 Backend

- FastAPI high-speed API

- Asynchronous processing
- Scalable design

10.3 Frontend

- Interactive dashboard
- Real-time predictions
- Visual risk indicators

11. Results & Output

- Accurate prediction of loan default probability
- Clear classification into risk categories
- Dynamic UI updates based on predictions
- Visualization using gauges and charts

12. Testing (UAT - User Acceptance Testing)

- Input validation tested
- API response verified
- UI correctly displays:
 - Probability
 - Risk level
 - Recommended action

UAT is completed successfully

13. Limitations

- Depends on dataset quality
- Limited real-world financial features
- No live database integration

14. Future Enhancements

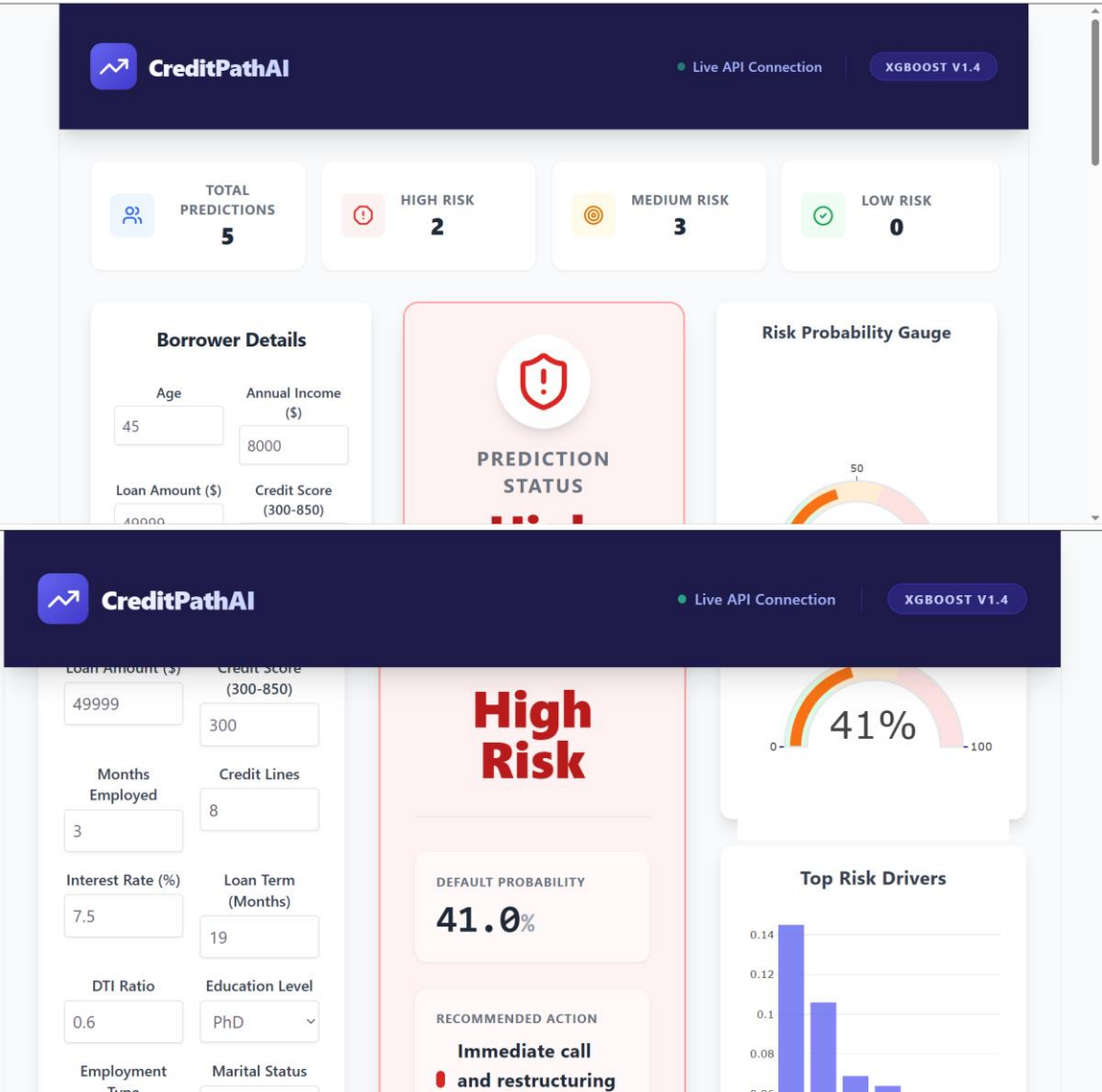
- Integration with real banking systems
- Use of deep learning models
- Adding real-time data streaming

- Mobile application development
- Explainable AI (SHAP/LIME)

15. Conclusion

CreditPathAI successfully demonstrates the integration of machine learning and full-stack development to solve a real-world financial problem. The system provides accurate risk predictions and actionable insights, making it useful for financial institutions in decision-making.

RESULTS:



CreditPathAI

● Live API Connection

XGBOOST V1.4

type

Unemploye

Divorced

Has Mortgage?

No

Has Dependents?

No

Loan Purpose

Other

Has Cosigner?

No

Run Risk Analysis

plan

High default probability detected based on historical data patterns. Urgent action required.

0.06

0.04

0.02

0

Age

L/I Ratio

Int. Rate

Int. Feature

Burden

Age Emp.

Cosigner

Dependents

CreditPathAI

● Live API Connection

XGBOOST V1.4

Recent Predictions

AGE	INCOME	RISK LEVEL	PROBABILITY
45 yrs	\$8,000	High Risk	40.96%
45 yrs	\$8,000	High Risk	40.96%
45 yrs	\$8,000	Medium Risk	33.23%
45 yrs	\$8,000	Medium Risk	35.52%
45 yrs	\$8,000	Medium Risk	29.56%

BACKEND:

CreditPathAI - Loan Default Prediction API

0.1.0 OAS 3.1

/openapi.json

default

- GET / Home
- POST /predict Predict

Schemas

HTTPValidationError > Expand all object

LoanRequest > Expand all object

ValidationError > Expand all object

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "age": 40,
    "income": 1500000,
    "loan_amount": 20000,
    "credit_score": 820,
    "months_employed": 120,
    "num_credit_lines": 8,
    "interest_rate": 5,
    "loan_term": 12,
    "dti_ratio": 0.05,
    "education": 3,
    "employment_type": 1,
    "marital_status": 1,
    "has_mortgage": 0,
    "has_dependents": 0,
    "loan_purpose": 1,
    "has_cosigner": 1
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code Details

200

Response body

```
{
  "probability": 0.1707,
  "risk": "Low Risk",
  "action": "Regular follow-up"
}
```

Download